

Backend Engineering Learning Progress & Execution Plan

Purpose:

This document is the execution and progress tracker for the Backend Engineering curriculum contained in the Excel curriculum file.

The Excel file defines **what needs to be learned**.

This document defines **how each topic will be completed, practiced, assessed, and tracked**.

1. Topic Completion Standard

Every topic should pass through the following stages:

1. **Lesson** — Study and understand the concept.
2. **Video** — Watch at least one relevant technical video where applicable.
3. **Practice** — Implement or experiment with the concept.
4. **Assignment** — Complete a practical engineering task.
5. **Assessment** — Answer questions and demonstrate understanding.
6. **Professional Ability** — Demonstrate what should be possible in a real engineering environment.
7. **Progress Review** — Record what was completed and identify the next topic/task.

A topic is considered **Complete** only when the learner has demonstrated the required professional ability, not merely watched or read the lesson.

2. Standard 2-Hour Session

Default session: 7:00 PM – 9:00 PM WAT

Time

Activity

7:00 – 7:25 PM	Lesson / Reading
7:25 – 7:45 PM	Video / Additional Resource
7:45 – 8:20 PM	Practical Exercise
8:20 – 8:45 PM	Assignment
8:45 – 8:55 PM	Assessment
8:55 – 9:00 PM	Progress update + Next Step

For difficult topics, the topic may be extended into a second session.

3. Progress Status

Use the following statuses:

- **Not Started**
 - **In Progress**
 - **Practiced**
 - **Assignment Submitted**
 - **Assessment Completed**
 - **Professional Ability Demonstrated**
 - **Completed**
 - **Needs Revision**
-

4. CURRICULUM EXECUTION PLAN

MODULE 1 — SOFTWARE DEVELOPMENT FOR BACKEND

Topic 1: Introduction to Software Engineering

Date: 27 August 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study the fundamentals of software engineering, including software development principles, engineering discipline, maintainability, reliability, quality, and the difference between simply writing code and engineering software.

Video:

Watch a beginner-to-intermediate software engineering fundamentals lecture.

Practice:

Take an existing backend project and identify areas related to maintainability, testing, documentation, architecture, and code quality.

Assignment:

Write a short engineering analysis of an existing backend application. Identify at least five engineering problems and propose improvements.

Assessment:

- What is software engineering?
- How is software engineering different from programming?
- What makes software maintainable?
- What factors determine software quality?
- Why is technical debt dangerous?

Professional Ability:

Can evaluate a backend application from an engineering perspective and identify improvements related to quality, maintainability, reliability, and architecture.

Status: Not Started

Topic 2: Specializations in Backend

Date: 28 August 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study backend engineering roles and specializations, including API engineering, database engineering, distributed systems, infrastructure, security, platform engineering, and application development.

Practice:

Map the responsibilities of a backend engineer across a real production system.

Assignment:

Create a backend engineering responsibility matrix showing what a junior, mid-level, senior, and staff backend engineer should typically handle.

Assessment:

- What does a backend engineer do?
- What is platform engineering?
- What is database engineering?
- What distinguishes a senior backend engineer from a junior engineer?

Professional Ability:

Can identify backend engineering responsibilities and understand how backend specializations contribute to a production system.

Status: Not Started

Topic 3: Software Development Life Cycle

Date: 29 August 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study requirements gathering, planning, design, implementation, testing, deployment, monitoring, maintenance, and iteration.

Practice:

Map the SDLC of a real application you have built.

Assignment:

Design the complete SDLC for a hypothetical fintech backend.

Assessment:

- What are the major stages of the SDLC?
- Why should requirements be defined before implementation?
- Where does testing happen?
- Where does deployment fit?
- Why is maintenance part of software development?

Professional Ability:

Can participate in and explain the complete lifecycle of a production software system.

Status: Not Started

Topic 4: Programming Languages

Date: 30 August 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Understand programming language concepts, syntax, types, memory, execution models, paradigms, and language selection.

Practice:

Implement the same small backend problem in two different languages.

Assignment:

Compare PHP, JavaScript/TypeScript, Java, and Python for backend development.

Assessment:

- What is a programming language?
- Compiled vs interpreted?
- Static vs dynamic typing?
- What factors influence language selection?

Professional Ability:

Can select and justify an appropriate programming language for a backend project.

Status: Not Started

Topic 5: Software Design

Date: 31 August 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study modularity, abstraction, encapsulation, separation of concerns, cohesion, coupling, SOLID principles, and maintainable design.

Practice:

Refactor a poorly structured backend module.

Assignment:

Design and implement a clean backend service following SOLID and separation-of-concern principles.

Assessment:

- What is coupling?
- What is cohesion?
- Explain SOLID.
- Why is abstraction useful?
- What makes a design maintainable?

Professional Ability:

Can design backend components that are modular, maintainable, testable, and extensible.

Status: Not Started

Topic 6: Testing

Date: 1 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study unit testing, integration testing, acceptance testing, regression testing, test automation, and testing strategies.

Practice:

Write tests for an existing API endpoint.

Assignment:

Build a backend feature with unit and integration tests.

Assessment:

- Unit vs integration testing?
- What is regression testing?
- What should be mocked?
- What makes a good test?

Professional Ability:

Can design and implement an effective automated testing strategy for backend software.

Status: Not Started

MODULE 2 — NETWORKING FUNDAMENTALS FOR BACKEND

Topic 7: Introduction to Networking

Date: 2 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Learn IP addresses, ports, clients, servers, packets, routing, and the basic principles of computer networking.

Practice:

Use networking tools such as ping, traceroute, curl, netstat/ss, and DNS utilities.

Assignment:

Document what happens when a user enters a website address into a browser.

Assessment:

- What is an IP address?
- What is a port?
- What is a packet?
- Client vs server?
- What does a router do?

Professional Ability:

Can troubleshoot basic network connectivity issues affecting backend applications.

Status: Not Started

Topic 8: DNS

Date: 3 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study domains, DNS records, resolvers, recursive queries, authoritative servers, A, AAAA, CNAME, MX, TXT, and TTL.

Practice:

Inspect DNS records for a real domain.

Assignment:

Configure a domain and explain its DNS resolution process.

Assessment:

Explain the complete process of resolving [api.example.com](#).

Professional Ability:

Can diagnose common DNS-related backend and deployment problems.

Status: Not Started

Topic 9: Network Protocols — TCP / UDP

Date: 4 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Understand TCP and UDP, connections, reliability, ordering, retransmission, ports, and use cases.

Practice:

Inspect TCP connections on a development machine.

Assignment:

Explain when a backend system should use TCP versus UDP.

Professional Ability:

Can select and reason about appropriate transport protocols for backend communication.

Status: Not Started

Topic 10: HTTP

Date: 5 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study HTTP requests, responses, methods, headers, status codes, cookies, caching, content negotiation, and HTTPS.

Practice:

Inspect API requests using browser developer tools and curl.

Assignment:

Design an HTTP API and document its requests and responses.

Professional Ability:

Can diagnose and design HTTP communication between clients and backend systems.

Status: Not Started

Topic 11: Sockets

Date: 6 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Understand sockets, connections, ports, TCP sockets, UDP sockets, and persistent communication.

Practice:

Build a basic client-server socket application.

Assignment:

Create a simple real-time communication server.

Professional Ability:

Can explain and implement low-level network communication using sockets.

Status: Not Started

Topic 12: Network Security

Date: 7 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study TLS/SSL, certificates, secure communication, encryption, authentication, and common network attacks.

Practice:

Inspect an HTTPS certificate and TLS connection.

Assignment:

Secure a backend API using HTTPS and document the security mechanisms involved.

Professional Ability:

Can identify and address common network security risks in backend systems.

Status: Not Started

Topic 13: Network Performance and Scalability

Date: 8 September 2026

Time: 7:00 PM – 9:00 PM

Lesson:

Study latency, bandwidth, throughput, connection pooling, timeouts, retries, and network bottlenecks.

Practice:

Measure API response times under different conditions.

Assignment:

Analyze and optimize a slow API request.

Professional Ability:

Can diagnose network-related performance bottlenecks.

Status: Not Started

MODULE 3 — FUNDAMENTALS OF DATABASES

Topic 14: Introduction to Databases

Date: 9 September 2026

Time: 7:00 PM – 9:00 PM

Study relational and non-relational databases, database servers, tables, documents, records, queries, and transactions.

Practice: Create a small database-backed application.

Assignment: Compare PostgreSQL, MySQL, MongoDB, and Redis and explain appropriate use cases.

Assessment: Explain when a relational database should be preferred over NoSQL.

Professional Ability: Can select an appropriate database technology for a backend system.

Status: Not Started

Topic 15: Data Modelling

Date: 10 September 2026

Time: 7:00 PM – 9:00 PM

Study entities, attributes, relationships, primary keys, foreign keys, normalization, denormalization, and schema design.

Practice: Design a database schema for an e-commerce application.

Assignment: Produce an ERD and implement the schema.

Professional Ability: Can design normalized and production-ready relational database schemas.

Status: Not Started

Topic 16: SQL / NoSQL / GraphDB / Key-Value / Wide-Column Databases

Date: 11 September 2026

Time: 7:00 PM – 9:00 PM

Study the major database categories and their strengths, limitations, and use cases.

Practice: Implement the same data model using different database paradigms.

Assignment: Choose database technologies for five different backend scenarios and justify each decision.

Professional Ability: Can make informed database technology decisions based on application requirements.

Status: Not Started

Topic 17: Indexing and Query Optimization

Date: 12 September 2026

Time: 7:00 PM – 9:00 PM

Study indexes, B-trees, composite indexes, query plans, full-table scans, selectivity, and optimization.

Practice: Create a large dataset and compare indexed and non-indexed queries.

Assignment: Diagnose and optimize at least three slow database queries.

Assessment: Explain why an index can improve a query but also negatively affect writes.

Professional Ability: Can diagnose database performance problems and optimize queries using appropriate indexes.

Status: Not Started

Topic 18: Database Transactions — ACID

Date: 13 September 2026

Time: 7:00 PM – 9:00 PM

Study atomicity, consistency, isolation, durability, commits, rollbacks, and transaction boundaries.

Practice: Implement a money-transfer transaction.

Assignment: Build a transaction-safe payment operation and test failure scenarios.

Professional Ability: Can correctly use database transactions for operations requiring consistency.

Status: Not Started

Topic 19: Database Security

Date: 14 September 2026

Time: 7:00 PM – 9:00 PM

Study authentication, authorization, SQL injection, least privilege, encryption, secrets, backups, and auditing.

Practice: Identify vulnerabilities in a deliberately insecure API.

Assignment: Harden a database-backed application against common database security threats.

Professional Ability: Can implement secure database access and protect sensitive data.

Status: Not Started

Topic 20: Database Internals

Date: 15 September 2026

Time: 7:00 PM – 9:00 PM

Study storage engines, pages, buffers, indexes, query execution, WAL/logging, and basic database architecture.

Practice: Investigate how PostgreSQL or MySQL executes a query.

Assignment: Explain the internal lifecycle of a database query from request to result.

Professional Ability: Can reason about database behavior beyond ORM-level abstractions.

Status: Not Started

MODULE 4 — DEPLOYMENT AND OPERATIONS

Topic 21: Introduction to Deployment

Date: 16 September 2026

Time: 7:00 PM – 9:00 PM

Learn build, release, deployment, environments, configuration, rollback, and production releases.

Practice: Deploy a backend application.

Assignment: Create a complete deployment process for a backend API.

Professional Ability: Can deploy a backend application safely to a production-like environment.

Topic 22: Deployment Strategies

Date: 17 September 2026

Time: 7:00 PM – 9:00 PM

Study rolling, blue-green, canary, recreate, and zero-downtime deployment strategies.

Assignment: Design a zero-downtime deployment strategy.

Professional Ability: Can choose an appropriate deployment strategy based on system requirements.

Topic 23: Deployment Automation

Date: 18 September 2026

Time: 7:00 PM – 9:00 PM

Study CI/CD, pipelines, automated testing, builds, releases, and deployment automation.

Assignment: Build a CI/CD pipeline for a backend project.

Professional Ability: Can automate backend build, test, and deployment processes.

Topic 24: Monitoring and Logging

Date: 19 September 2026

Time: 7:00 PM – 9:00 PM

Study application logs, metrics, tracing, alerts, dashboards, health checks, and observability.

Assignment: Add structured logging and monitoring to a backend application.

Professional Ability: Can monitor a production backend and investigate operational issues.

Topic 25: Incident Response

Date: 20 September 2026

Time: 7:00 PM – 9:00 PM

Study incident detection, triage, mitigation, root-cause analysis, communication, and postmortems.

Assignment: Conduct a simulated production outage and write a postmortem.

Professional Ability: Can respond systematically to production incidents.

Topic 26: Operations Best Practices

Date: 21 September 2026

Time: 7:00 PM – 9:00 PM

Study infrastructure management, reliability, backups, security, observability, automation, and SRE principles.

Assignment: Produce an operational checklist for a production backend system.

Professional Ability: Can operate and maintain a backend system reliably.

MODULE 5 — TESTING

Topic 27: Introduction to Testing

Date: 22 September 2026 | **Time:** 7:00 PM – 9:00 PM

Learn testing fundamentals, test levels, test cases, assertions, test environments, and test strategy.

Practice: Add tests to an existing API.

Assignment: Create a testing strategy for a backend application.

Professional Ability: Can determine what and how a backend system should be tested.

Topic 28: Test-Driven Development

Date: 23 September 2026 | **Time:** 7:00 PM – 9:00 PM

Study Red-Green-Refactor and practical TDD.

Assignment: Build a backend feature using TDD.

Professional Ability: Can use automated tests to drive reliable implementation.

Topic 29: Debugging and Troubleshooting

Date: 24 September 2026 | **Time:** 7:00 PM – 9:00 PM

Learn systematic debugging, logs, stack traces, reproduction, isolation, profiling, and root-cause analysis.

Assignment: Diagnose several deliberately introduced backend bugs.

Professional Ability: Can systematically identify and resolve backend failures.

Topic 30: Continuous Integration

Date: 25 September 2026 | **Time:** 7:00 PM – 9:00 PM

Study automated testing and quality checks inside CI pipelines.

Assignment: Configure CI to run linting, tests, and builds.

Professional Ability: Can integrate automated quality checks into software delivery.

Topic 31: Testing Pyramid

Date: 26 September 2026 | **Time:** 7:00 PM – 9:00 PM

Study unit, integration, and end-to-end testing and how to balance them.

Assignment: Review an existing test suite and restructure it according to the testing pyramid.

Professional Ability: Can design a balanced testing strategy.

MODULE 6 — API DESIGN AND IMPLEMENTATION

Topic 32: Introduction to APIs

Date: 27 September 2026 | **Time:** 7:00 PM – 9:00 PM

Understand APIs, clients, servers, contracts, endpoints, requests, responses, and API lifecycle.

Assignment: Design a complete API for a simple business application.

Professional Ability: Can design clear and maintainable APIs.

Topic 33: RESTful API Design

Date: 28 September 2026 | **Time:** 7:00 PM – 9:00 PM

Study resources, HTTP verbs, status codes, statelessness, pagination, filtering, sorting, versioning, and idempotency.

Assignment: Build a production-quality REST API.

Professional Ability: Can design REST APIs following sound engineering principles.

Topic 34: API Documentation

Date: 29 September 2026 | **Time:** 7:00 PM – 9:00 PM

Study OpenAPI/Swagger and effective API documentation.

Assignment: Fully document an existing backend API.

Professional Ability: Can produce documentation that allows another developer to consume an API without assistance.

Topic 35: HTTP

Date: 30 September 2026 | **Time:** 7:00 PM – 9:00 PM

Deep dive into HTTP semantics, headers, caching, authentication, cookies, content types, and status codes.

Professional Ability: Can diagnose and correctly implement HTTP-based backend communication.

Topic 36: RPC / gRPC

Date: 1 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study RPC, protobuf, gRPC, streaming, and service-to-service communication.

Assignment: Build two backend services communicating using gRPC.

Professional Ability: Can determine when RPC/gRPC is preferable to REST.

Topic 37: API Security

Date: 2 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study authentication, authorization, JWT, OAuth, rate limiting, input validation, CORS, CSRF, and API abuse.

Assignment: Secure an API against common attacks.

Professional Ability: Can design and implement secure APIs.

Topic 38: API Performance and Scalability

Date: 3 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study caching, pagination, connection pooling, asynchronous processing, rate limiting, and horizontal scaling.

Assignment: Optimize a slow API and demonstrate measurable performance improvement.

Professional Ability: Can identify and resolve API performance bottlenecks.

Topic 39: API Case Studies

Date: 4 October 2026 | **Time:** 7:00 PM – 9:00 PM

Analyze real-world APIs and their design decisions.

Assignment: Perform a technical review of a major public API.

Professional Ability: Can critically evaluate real-world API architectures.

MODULE 7 — NETWORKING CONCEPTS FOR BACKEND ENGINEERING

Topics 40–48

40. TCP — 5 October 2026

Practice TCP connections and explain connection establishment and termination.

41. UDP — 6 October 2026

Compare UDP and TCP and implement a basic UDP communication example.

42. Forward Proxies — 7 October 2026

Configure and explain a forward proxy.

43. Reverse Proxies — 8 October 2026

Configure Nginx as a reverse proxy for a backend application.

44. Layer 7 Proxies — 9 October 2026

Study application-aware routing and HTTP-level proxying.

45. Layer 4 Proxies — 10 October 2026

Study transport-level proxying and compare it with Layer 7.

46. DNS — 11 October 2026

Troubleshoot DNS resolution and production domain configuration.

47. Load Balancing — 12 October 2026

Implement and test load balancing between multiple backend instances.

48. Networking Case Study — 13 October 2026

Analyze the networking architecture of a production-scale application.

Professional Ability for Module:

Can understand, configure, troubleshoot, and reason about the networking infrastructure supporting backend systems.

MODULE 8 — PLANNING AND DOCUMENTING SOFTWARE PROCESS

Topic 49: Requirements Gathering

Date: 14 October 2026 | **Time:** 7:00 PM – 9:00 PM

Practice extracting functional and non-functional requirements from a business problem.

Assignment: Produce requirements for a production backend system.

Professional Ability: Can translate business requirements into actionable engineering requirements.

Topic 50: Asking the Right Questions

Date: 15 October 2026 | **Time:** 7:00 PM – 9:00 PM

Learn how engineers clarify ambiguous requirements.

Assignment: Analyze an incomplete product requirement and produce the questions required before development.

Professional Ability: Can identify ambiguity and prevent engineering teams from building the wrong system.

Topic 51: Back-of-the-Envelope Estimation

Date: 16 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study traffic, storage, bandwidth, requests per second, concurrency, and capacity estimation.

Assignment: Estimate infrastructure requirements for a system serving one million users.

Professional Ability: Can perform basic system capacity estimates before architecture decisions.

Topic 52: Software Design

Date: 17 October 2026 | **Time:** 7:00 PM – 9:00 PM

Create component-level software designs.

Assignment: Produce a detailed software design for a backend feature.

Topic 53: Project Planning

Date: 18 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study milestones, dependencies, risks, estimation, sequencing, and delivery planning.

Assignment: Create an engineering delivery plan.

Topic 54: Software Documentation

Date: 19 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study technical documentation, API documentation, architecture documentation, and operational documentation.

Assignment: Document an existing backend system.

Topic 55: Technical Decision Document (TDD)

Date: 20 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study how technical decisions are recorded and justified.

Assignment: Write three technical decision documents for a backend project.

Topic 56: Architecture Decision Record (ADR)

Date: 21 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study ADR structure and decision management.

Assignment: Create ADRs for database, caching, authentication, and deployment decisions.

Topic 57: Architecture Design Document

Date: 22 October 2026 | **Time:** 7:00 PM – 9:00 PM

Create a complete architecture design document.

Professional Ability: Can communicate architecture decisions clearly to engineers, product stakeholders, and technical leadership.

MODULE 9 — MICROSERVICES

Topic 58: Introduction to Microservices

Date: 23 October 2026 | **Time:** 7:00 PM – 9:00 PM

Understand monoliths, modular monoliths, microservices, service boundaries, and trade-offs.

Assignment: Determine whether a hypothetical application should remain a monolith or become microservices.

Professional Ability: Can make informed architectural decisions regarding service decomposition.

Topic 59: Designing Microservices

Date: 24 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study bounded contexts, service ownership, data ownership, and service boundaries.

Assignment: Decompose an e-commerce platform into appropriate services.

Topic 60: Implementing Microservices

Date: 25 October 2026 | **Time:** 7:00 PM – 9:00 PM

Build multiple independent backend services.

Assignment: Implement a small microservices system.

Topic 61: Communicating Between Microservices

Date: 26 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study REST, gRPC, messaging, queues, events, synchronous and asynchronous communication.

Assignment: Implement service-to-service communication using both synchronous and asynchronous approaches.

Topic 62: Managing Microservices

Date: 27 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study service discovery, configuration, deployment, monitoring, logging, and versioning.

Professional Ability: Can operate a multi-service backend architecture.

Topic 63: Security in Microservices

Date: 28 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study service authentication, authorization, secrets, network security, and trust boundaries.

Assignment: Secure communication between multiple backend services.

Topic 64: Microservices Case Studies

Date: 29 October 2026 | **Time:** 7:00 PM – 9:00 PM

Analyze real-world microservice architectures and identify their trade-offs.

MODULE 10 — SYSTEM DESIGN FOR BACKEND SYSTEMS

Topic 65: System Design Introduction

Date: 30 October 2026 | **Time:** 7:00 PM – 9:00 PM

Learn how to approach system design problems from requirements through architecture.

Assignment: Design a scalable backend for a social media application.

Professional Ability: Can approach an unfamiliar system design problem systematically.

Topic 66: Design Principles

Date: 31 October 2026 | **Time:** 7:00 PM – 9:00 PM

Study scalability, reliability, maintainability, availability, consistency, and simplicity.

Topic 67: Architectural Patterns

Date: 1 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study layered architecture, event-driven architecture, microservices, modular monoliths, CQRS, and other patterns.

Assignment: Compare architectures for three real-world applications.

Topic 68: Designing for Scalability and Performance

Date: 2 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study caching, load balancing, horizontal scaling, queues, database optimization, and CDN usage.

Assignment: Scale a backend application from 1,000 to 1,000,000 users.

Topic 69: Designing for Reliability and Availability

Date: 3 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study redundancy, failover, health checks, retries, circuit breakers, backups, and disaster recovery.

Assignment: Design a highly available backend system.

Topic 70: Designing for Security

Date: 4 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study authentication, authorization, encryption, network security, secrets, threat modelling, and secure architecture.

Assignment: Perform a security review of a system design.

Topic 71: System Design Case Studies

Date: 5 November 2026 | **Time:** 7:00 PM – 9:00 PM

Complete real-world design problems such as:

- URL shortener
- Chat system
- Payment system
- Ride-hailing system
- E-commerce platform
- Notification system

- File storage system

Professional Ability: Can independently conduct a structured system-design interview and produce a defensible production architecture.

MODULE 11 — DISTRIBUTED SYSTEMS

Topic 72: Distributed Systems — Fundamental Concepts

Date: 6 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study distributed nodes, coordination, communication, consistency, availability, failures, and distributed state.

Assignment: Explain why distributed systems are fundamentally harder than single-server systems.

Topic 73: Communication in Distributed Systems

Date: 7 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study messaging, RPC, queues, event streaming, retries, timeouts, and delivery guarantees.

Assignment: Design communication between multiple distributed services.

Topic 74: Distributed Coordination

Date: 8 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study leader election, distributed locks, consensus, coordination services, and race conditions.

Assignment: Design a distributed locking mechanism and explain its failure scenarios.

Topic 75: Fault Tolerance and Reliability

Date: 9 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study failure detection, retries, timeouts, circuit breakers, redundancy, replication, and recovery.

Assignment: Design a fault-tolerant service.

Professional Ability: Can design systems that continue operating despite partial failures.

Topic 76: Distributed Scalability

Date: 10 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study horizontal scaling, partitioning, load balancing, replication, and distributed workloads.

Topic 77: Distributed Systems Security

Date: 11 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study authentication, authorization, secure communication, service identity, secrets, and trust boundaries.

Topic 78: Distributed Systems Design Patterns

Date: 12 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study event-driven architecture, saga, outbox, CQRS, circuit breaker, bulkhead, retry, and idempotency patterns.

Assignment: Select appropriate distributed-system patterns for a real-world system.

Topic 79: Distributed Systems Case Studies

Date: 13 November 2026 | **Time:** 7:00 PM – 9:00 PM

Analyze distributed databases, distributed storage systems, large-scale messaging systems, and production architectures.

Professional Ability: Can reason about the architecture, trade-offs, and failure modes of large distributed systems.

MODULE 12 — ADVANCED DATABASE CONCEPTS

Topic 80: Database Engines

Date: 14 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study storage engines, execution engines, memory, caching, transactions, and internal architecture.

Assignment: Compare the internal architecture of PostgreSQL and MySQL.

Topic 81: Advanced Indexing and Query Optimization

Date: 15 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study execution plans, composite indexes, covering indexes, selectivity, query planners, and advanced optimization.

Assignment: Optimize a database workload containing several million records.

Professional Ability: Can perform advanced database performance analysis.

Topic 82: Concurrency Control in Databases

Date: 16 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study locks, isolation levels, MVCC, deadlocks, race conditions, and concurrent transactions.

Assignment: Reproduce and resolve a database race condition.

Professional Ability: Can design transaction workflows that remain correct under concurrent access.

Topic 83: Database Partitioning

Date: 17 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study horizontal and vertical partitioning, partition keys, range partitioning, list partitioning, and operational trade-offs.

Assignment: Partition a large database table and measure the impact.

Professional Ability: Can determine when and how database partitioning should be introduced.

Topic 84: Database Replication

Date: 18 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study primary/replica architectures, replication lag, read replicas, failover, and consistency.

Assignment: Design a replicated database architecture for a high-traffic backend.

Professional Ability: Can design database replication strategies for availability and read scalability.

Topic 85: Database Sharding

Date: 19 November 2026 | **Time:** 7:00 PM – 9:00 PM

Study horizontal sharding, shard keys, routing, rebalancing, cross-shard queries, and operational complexity.

Assignment: Design a sharded database architecture capable of supporting a very large dataset.

Professional Ability: Can evaluate when sharding is necessary and design a reasonable sharding strategy.

Topic 86: Database Applications in the Real World

Date: 20 November 2026

Time: 7:00 PM – 9:00 PM

Study real-world database architectures and trade-offs.

Assignment: Design the complete database architecture for a production-scale application incorporating:

- Data modelling
- Indexing
- Transactions
- Caching
- Replication
- Partitioning
- Sharding
- Backup
- Security
- Monitoring

Professional Ability: Can design and justify a production-grade database architecture based on real business and technical requirements.

5. FINAL CAPSTONE PROJECT

After completing the curriculum, build one production-grade backend system that combines the major concepts learned.

Recommended Project

High-Scale Transaction & Logistics Platform

The system should include:

- REST API
- Authentication and authorization
- PostgreSQL/MySQL
- Redis caching
- Background queues
- Notifications
- API rate limiting
- Database transactions
- Indexing and query optimization
- Automated testing
- Docker
- CI/CD
- Monitoring and logging
- Reverse proxy
- Load balancing
- Horizontal scaling
- Microservices or modular architecture
- Event-driven communication
- Database replication
- Database partitioning
- Security controls
- Architecture documentation
- ADRs
- System design documentation

Capstone Deliverables

1. Product requirements
 2. System requirements
 3. Architecture diagram
 4. Database ERD
 5. API documentation
 6. Technical decision documents
 7. Architecture Decision Records
 8. Source code
 9. Automated tests
 10. CI/CD pipeline
 11. Deployment
 12. Monitoring
 13. Security review
 14. Performance test
 15. Load test
 16. Incident simulation
 17. Final technical presentation
-

6. FINAL PROFESSIONAL ABILITY

Upon completing the curriculum and capstone, the learner should be able to:

Design, build, test, deploy, operate, troubleshoot, secure, and scale production-grade backend systems while making and communicating sound engineering and architectural decisions.

The learner should also be capable of:

- Working independently on complex backend systems
- Participating effectively in architecture discussions
- Performing system design
- Diagnosing production issues
- Optimizing databases
- Designing APIs
- Working with distributed systems
- Designing scalable architectures
- Making technology decisions based on requirements

- Communicating technical decisions professionally
 - Reviewing other engineers' designs and implementations
-

7. PROGRESS REVIEW TEMPLATE

At the end of every topic, record:

Topic:

Date Completed:

Time Spent:

Lesson: Completed / Not Completed

Video: Completed / Not Completed

Practice: Completed / Not Completed

Assignment: Completed / Not Completed

Assessment Score: ____ / ____

Professional Ability: Demonstrated / Not Demonstrated

What I Learned:

What I Struggled With:

What I Need to Revise:

Evidence / Project Link:

Overall Topic Status:

Next Step:

8. DAILY NEXT-STEP FORMAT

At the end of every session, the following must be recorded:

Completed Today:

[What was actually completed]

Outstanding:

[What remains incomplete]

Revision Required:

[Concepts that need another review]

Next Topic:

[Next topic from the curriculum]

Next Session Date:

[Date]

Next Session Time:

[Time]

Expected Outcome:

[What must be achieved before the next session is considered complete]

9. IMPORTANT RULE

Completion is based on **demonstrated ability**, not consumption of content.

Watching a video does not mean the topic is complete.

Reading a lesson does not mean the topic is complete.

Passing an assessment alone does not mean the topic is complete.

A topic is complete when the learner can **explain the concept, apply it in practice, complete the assignment, pass the assessment, and demonstrate the corresponding professional ability.**